

AIMO Proof Pilot — Final Grade

Problem (#_ID): 1_DIVALL

Submission: team_4

Scoring scale (0/1/6/7) — two binary decisions:

Step 1. Is the solution essentially complete?

- If **no**: award **1** for significant partial progress, otherwise **0**.
- If **yes**: award **7**, or **6** if there is a minor error or omission.

Grade against the **markscheme** (docs/markschemes.pdf).

1. Final mark

Final mark (choose one):

2. Reasoning

Justify the mark with explicit reference to the markscheme points (e.g. **ME1**) where possible.

Complete solution: it proves the a_i share a common residue r with $0 < r < g$, eliminates $k \geq 4$, bounds $g \leq 998285$, and verifies that this bound is attained. Complete and correct.